

Beyond the Bethe Approximation of the Permanent

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Abstract

The canonical Bethe approximation gives a deterministic approximation to the permanent of every nonnegative matrix within a factor of $(\sqrt{2})^n$. We improve the base of this exponential factor: for some absolute constant $c < \sqrt{2}$, there is a deterministic polynomial-time c^n -approximation for the permanent of every nonnegative matrix. This shows that the canonical Bethe guarantee is not a barrier for deterministic approximation of the permanent. The proof augments the Bethe lower bound with a new certificate tailored to matrices on which that lower bound loses nearly the full factor.

The author supplied the high-level plan of attack, and the proof was developed in an interaction with ChatGPT 5.6 Sol Pro. The author subsequently verified the results. Codex assisted with proof checking, manuscript assembly, and typesetting.

1 Introduction

The permanent of a matrix $A \in \mathbb{R}_{\geq 0}^{n \times n}$ is

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n A_{i,\sigma(i)}.$$

Here S_n is the set of permutations of $\{1, \dots, n\}$. Computing or approximating the permanent is one of the canonical problems in theoretical computer science and the field of counting and sampling. Valiant proved that computing the permanent is #P-hard even for 0/1 matrices [Val79]. On the other hand, Jerrum, Sinclair, and Vigoda gave a randomized fully polynomial-time approximation scheme for every nonnegative matrix [JSV04]. The best general deterministic algorithms remain exponentially worse.

A particularly natural deterministic approximation is the *Bethe permanent*. It is the maximum of a concave function over the Birkhoff polytope, the set of doubly stochastic matrices, and can therefore be computed efficiently; we denote it by $\text{Bethe}(A)$. Gurvits proved that it is a lower bound for the permanent, building on Schrijver’s permanental inequality [Sch98; Gur11]; Vontobel developed its variational and graph-cover interpretations [Von13]. Anari and Rezaei proved the sharp universal sandwich

$$\text{Bethe}(A) \leq \text{per}(A) \leq 2^{n/2} \text{Bethe}(A)$$

^{*}The accompanying Lean 4 formalization is available at <https://github.com/nimaanari/formalization-beyond-bethe>. The corresponding Palomar Registry entry is available at <https://palomar-registry.org/entry?id=PALOMAR-2026-08-29-000006&version=1>.

[AR24]. The factor $2^{n/2}$ is attained by a block diagonal matrix with 2×2 all-ones blocks. Thus no analysis of the Bethe permanent alone can improve the base $\sqrt{2}$.

This paper goes beyond that barrier by augmenting the Bethe certificate rather than replacing it. Our main result is the following.

Theorem 1 (Main theorem). *There is an absolute constant $c < \sqrt{2}$ and a deterministic polynomial-time algorithm that, given $A \in \mathbb{Q}_{\geq 0}^{n \times n}$, returns a number $L(A)$ satisfying*

$$L(A) \leq \text{per}(A) \leq c^n L(A).$$

We make no attempt to optimize c . The point is the uniform improvement in the base of the exponential approximation; optimizing the constants would obscure the argument.

Independent work. Independently, Narang and Perkins proved a similar result: a deterministic polynomial-time $(\sqrt{2} - \varepsilon)^n$ -approximation for some absolute constant $\varepsilon > 0$ [NP26]. Their approach also exploits nearly isolated weighted 2×2 blocks, through a structural correction to the Bethe approximation.

1.1 Proof overview

The proof couples a new lower-bound certificate with a stability analysis of the Bethe upper bound. For this overview, suppose that A is positive and let

$$\Delta(A) = \frac{n}{2} \log 2 + \log \text{Bethe}(A) - \log \text{per}(A)$$

be the slack in that upper bound. If $\Delta(A) = \Omega(n)$, then the Bethe permanent already beats $(\sqrt{2})^n$ by an exponential factor. We therefore only need a better certificate when $\Delta(A)$ is a small constant multiple of n . The argument has three main ideas.

A lower bound that remembers pairs of rows. A matching partitions the rows into pairs and unmatched singletons. A singleton contributes a linear polynomial whose capacity is its contribution to the Bethe lower bound. For a pair of rows r, s , we instead use the quadratic polynomial

$$Q_{rs}(z) = \sum_{j < k} (A_{rj}A_{sk} + A_{rk}A_{sj}) z_j z_k.$$

The coefficient of $z_j z_k$ is the total weight of the two ways to assign rows r, s to the distinct columns j, k . We multiply the polynomials of all row clusters and pair the result with another polynomial that enforces that every column is used exactly once. A coefficient inequality for real stable polynomials—a class defined in Section 2—then turns this encoding into a lower bound for $\text{per}(A)$ [AO17]. On an exact 2×2 all-ones block, the paired cluster recovers the factor 2 lost by the Bethe permanent.

Near equality forces approximate 2×2 blocks. Sample a permutation σ with probability proportional to its weight $\prod_i A_{i, \sigma(i)}$, and let $P_{ij} = \mathbb{P}_\sigma[\sigma(i) = j]$. Thus row i of P records the marginal distribution of the column assigned to row i . Anari and Rezaei analyze the following sequential experiment. Order the rows uniformly at random and process them in that order; when row i is

reached, choose an unused column with probability proportional to the corresponding entries of P_i . This experiment produces another distribution on permutations. Their upper bound compares it with the original weighted distribution and then applies a one-row inequality.

We keep the information lost in both steps. This gives an exact decomposition of $\Delta(A)$ into three nonnegative contributions: the suboptimality of P in the Bethe program, the sum of the deficits in the one-row inequality, and the average relative entropy between the true and sequential distributions. Consequently, when $\Delta(A)$ is small, all three contributions are small.

The one-row inequality is stable: a row with small deficit must be close to a probability vector supported uniformly on two columns. Hence most rows of P have two entries near $1/2$. These large entries form a bipartite graph of maximum degree two. The relative-entropy contribution rules out many long cycles in this graph, so most rows lie in approximate copies of $K_{2,2}$, the complete bipartite graph on two rows and two columns. This is the rigidity statement behind the improvement.

Transfer the blocks and collect their gain. The marginal matrix P is not available to the algorithm. We optimize a slightly regularized Bethe objective instead, adding a small multiple of the sum of the Shannon entropies of its rows. The unique optimizer X is positive. Its Karush–Kuhn–Tucker optimality equations give positive row and column scaling factors r_i, c_j such that

$$A_{ij} = r_i c_j X_{ij}^{1+\tau} (1 - X_{ij}).$$

Here τ is a small positive regularization parameter. A global identity derived from this factorization measures how much the relevant coordinates can change between P and X . It shows that most of the approximate $K_{2,2}$'s found in P remain useful at X . On each such block, the paired certificate improves on the two singleton certificates by a constant factor. A maximum-weight row matching collects these gains on a linear number of disjoint blocks, improving the lower bound by $e^{\Omega(n)}$.

The key point is the interaction between the two sides of the argument. The stability analysis identifies precisely where the Bethe bound can be nearly tight, while the paired certificate is designed to recover weight on those same configurations.

1.2 Related work

The Bethe permanent was introduced and analyzed systematically by Vontobel [Von13]. Its lower-bound property follows from Schrijver's inequality and Gurvits's variational formulation [Sch98; Gur11]. Anari and Rezaei proved the matching $2^{n/2}$ upper bound [AR24]. Our proof begins with their sequential-distribution argument but uses its discarded relative-entropy term as a structural statistic.

Graph covers give another interpretation of the Bethe permanent. Ng and Vontobel studied the role of double covers [NV22]; Huang, Kashyap, and Vontobel developed degree- M Bethe and Sinkhorn permanent bounds [HKV24]. These works analyze the Bethe approximation and its finite-cover counterparts. Our certificate instead recovers weight that the Bethe objective necessarily loses on 2×2 blocks.

The stable-polynomial input is the multiaffine coefficient inequality of Anari and Oveis Gharan [AO17]. It proves one-sidedness of each paired certificate; the global argument deciding which pairs are profitable is information-theoretic.

Organization. Section 2 records notation, the Bethe bound, and the two stable-polynomial facts used in the proof. Section 3 constructs the paired lower certificate. Sections 4 and 5 prove the exact slack identity and the stability of its row term. Section 6 extracts the robust $K_{2,2}$ structure. Sections 7 and 8 transfer this structure to the regularized optimizer and prove the paired gain. Section 9 combines the estimates. Section A treats zero entries and finite-precision computation.

2 Preliminaries

Notation. We write $[n] = \{1, \dots, n\}$ and let S_n denote the set of permutations of $[n]$. The Birkhoff polytope is

$$\mathcal{B}_n = \left\{ X \in \mathbb{R}_{\geq 0}^{n \times n} \mid \sum_j X_{ij} = 1 \ \forall i, \sum_i X_{ij} = 1 \ \forall j \right\}.$$

For a probability vector p , its Shannon entropy is

$$H(p) = - \sum_j p_j \log p_j,$$

with the convention $0 \log 0 = 0$. We write $H_2(t) = -t \log t - (1-t) \log(1-t)$ for binary entropy. For probability measures μ, ν on the same finite set, their Kullback–Leibler (KL) divergence is

$$D_{\text{KL}}(\mu \parallel \nu) = \sum_x \mu(x) \log \frac{\mu(x)}{\nu(x)}.$$

All logarithms are natural. We use the same formula to write $H(\mu)$ for the entropy of a finite probability measure, $H(Z)$ for the entropy of a finite-valued random variable, and $H(Z \mid W)$ for conditional entropy.

2.1 The Bethe permanent

For $A \in \mathbb{R}_{\geq 0}^{n \times n}$ and $X \in \mathcal{B}_n$, define the Bethe objective

$$\beta_A(X) = \sum_{i,j} \left[X_{ij} \log \frac{A_{ij}}{X_{ij}} + (1 - X_{ij}) \log(1 - X_{ij}) \right]. \quad (1)$$

At $X_{ij} \in \{0, 1\}$, we interpret the summands by continuity. The Bethe permanent is

$$\text{Bethe}(A) = \exp \left(\max_{X \in \mathcal{B}_n} \beta_A(X) \right).$$

For a nonnegative matrix A , we use the same definition with the convention that $X_{ij} \log(A_{ij}/X_{ij}) = -\infty$ when $A_{ij} = 0 < X_{ij}$, and that this term is zero when $X_{ij} = 0$. Equivalently, this is the limit of $\text{Bethe}(A + \delta J)$ as $\delta \downarrow 0$, where J is the all-ones matrix. Vontobel proved that the objective is concave on $\mathcal{B}_n$ [Von13]. The following sharp sandwich combines the lower bound of Gurvits with the upper bound of Anari and Rezaei [Gur11; AR24].

Theorem 2 (Bethe sandwich). *For every $A \in \mathbb{R}_{\geq 0}^{n \times n}$,*

$$\text{Bethe}(A) \leq \text{per}(A) \leq 2^{n/2} \text{Bethe}(A). \quad (2)$$

Both inequalities are tight.

We first prove the result for positive matrices. Zero entries are treated by a quantitative smoothing argument in [Section A](#).

2.2 Capacity and real stability

We need one coefficient inequality for multivariate polynomials. Its variational quantity is the following scale-invariant lower envelope. Let $q \in \mathbb{R}_{\geq 0}[z_1, \dots, z_m]$ be homogeneous of degree d . For $\alpha \in \mathbb{R}_{\geq 0}^m$ with $\sum_j \alpha_j = d$, define its capacity at α by

$$\text{Cap}_\alpha(q) = \inf_{z \in \mathbb{R}_{>0}^m} \frac{q(z)}{z^\alpha}. \quad (3)$$

Here and below $z^\alpha = \prod_j z_j^{\alpha_j}$.

A polynomial is *multiaffine* if every variable has degree at most one. A polynomial $q \in \mathbb{R}[z_1, \dots, z_m]$ is *real stable* if $q(z_1, \dots, z_m) \neq 0$ whenever every z_j has positive imaginary part. Products of real stable polynomials in disjoint variable sets are real stable. We use the following two facts.

Lemma 3 (Quadratic stability criterion). *Let q be a nonzero homogeneous multiaffine quadratic with nonnegative coefficients. If the Hessian of q has at most one positive eigenvalue, then q is real stable.*

Proof. Write $q(z) = \frac{1}{2} z^\top H z$, where H is the Hessian. Suppose that $q(x + iy) = 0$ for some $x \in \mathbb{R}^m$ and $y \in \mathbb{R}_{>0}^m$. Separating real and imaginary parts gives

$$x^\top H x = y^\top H y, \quad x^\top H y = 0.$$

The nonnegative coefficients and $y > 0$ imply $y^\top H y = 2q(y) > 0$. Since H has at most one positive eigenvalue, its quadratic form is nonpositive on the H -orthogonal complement of y , namely on the vectors x satisfying $x^\top H y = 0$. Thus $x^\top H x \leq 0$, contradicting the first equality. This is the degree-two case of Brändén's half-plane-property criterion [[Brä07](#)]. $\square$

For a set $S \subseteq [m]$, write $z^S = \prod_{j \in S} z_j$. For multiaffine polynomials $p(z) = \sum_S p_S z^S$ and $q(z) = \sum_S q_S z^S$, write

$$\langle p, q \rangle = \sum_S p_S q_S.$$

Thus this inner product pairs coefficients of the same squarefree monomial.

Theorem 4 (Stable coefficient inequality). *Let $p, q \in \mathbb{R}_{\geq 0}[z_1, \dots, z_m]$ be homogeneous, multiaffine, real stable polynomials of the same degree d . For every $\alpha \in [0, 1]^m$ with $\sum_j \alpha_j = d$,*

$$\langle p, q \rangle \geq \prod_{j=1}^m \alpha_j^{\alpha_j} (1 - \alpha_j)^{1 - \alpha_j} \text{Cap}_\alpha(p) \text{Cap}_\alpha(q). \quad (4)$$

The conventions $0^0 = 1$ and $1^0 = 1$ are used at the boundary.

This is the multiaffine specialization of the coefficient inequality of Anari and Oveis Gharan [[AO17](#)]. The inner product in [Eq. \(4\)](#) pairs the same squarefree monomial in p and q , rather than complementary monomials.

We use the following elementary entropy certificate for capacity. Its one-sided form is exactly what the proof and the algorithm need.

Lemma 5 (Entropy certificate for capacity). *Let*

$$q(z) = \sum_{E \in \mathcal{E}} c_E z^E, \quad c_E > 0,$$

where $\mathcal{E} \subseteq \mathbb{Z}_{\geq 0}^m$ and $z^E = \prod_j z_j^{E_j}$. If θ is a probability distribution on $\mathcal{E}$ with $\sum_E \theta_E E = \alpha$, then

$$\sum_{E \in \mathcal{E}} \theta_E \log \frac{c_E}{\theta_E} \leq \log \text{Cap}_\alpha(q). \quad (5)$$

Terms with $\theta_E = 0$ are interpreted by continuity.

Proof. Fix a feasible θ . For every $z > 0$, the log-sum inequality gives

$$\begin{aligned} \log q(z) - \langle \alpha, \log z \rangle &\geq \sum_E \theta_E \log \frac{c_E z^E}{\theta_E} - \sum_E \theta_E \langle E, \log z \rangle \\ &= \sum_E \theta_E \log \frac{c_E}{\theta_E}. \end{aligned}$$

Taking the infimum over z proves the claim. $\square$

3 A paired lower certificate

Throughout the structural argument we assume $n \geq 2$; the case $n = 1$ is exact. We augment the Bethe lower bound by clustering disjoint pairs of rows. The construction is valid for every interior point of the Birkhoff polytope; later we evaluate it at the optimizer of a regularized Bethe objective.

Fix a positive matrix $A \in \mathbb{R}_{>0}^{n \times n}$ and an interior point $X \in \mathcal{B}_n$, meaning that every entry of X is positive. For a row i , define

$$S_i(A, X) = \prod_j \left(\frac{A_{ij}}{X_{ij}} \right)^{X_{ij}} (1 - X_{ij})^{1-X_{ij}}. \quad (6)$$

The product of the singleton factors is exactly the exponential of the Bethe objective:

$$\prod_i S_i(A, X) = e^{\beta A(X)}. \quad (7)$$

For two distinct rows r, s , let

$$Q_{rs}(z) = \sum_{j < k} (A_{rj} A_{sk} + A_{rk} A_{sj}) z_j z_k, \quad \alpha_j^{rs} = X_{rj} + X_{sj}. \quad (8)$$

The coefficient of $z_j z_k$ is the combined weight of the two assignments of rows r, s to columns j, k , while α_j^{rs} is the total mass that these two rows place on column j . The column constraints on X imply

$$0 < \alpha_j^{rs} \leq 1, \quad \sum_j \alpha_j^{rs} = 2.$$

Writing e_j for the j -th standard basis vector, the vectors in $[0, 1]^n$ whose coordinates sum to two form the convex hull of $e_j + e_k$, $j < k$. Thus α^{rs} lies in the convex hull of the exponent vectors of the monomials in Q_{rs} , as required for its capacity. Define

$$S_{rs}^{\text{pair}}(A, X) = \prod_j (1 - \alpha_j^{rs})^{1 - \alpha_j^{rs}} \text{Cap}_{\alpha^{rs}}(Q_{rs}), \quad \Gamma_{rs}(A, X) = \frac{S_{rs}^{\text{pair}}(A, X)}{S_r(A, X)S_s(A, X)}. \quad (9)$$

If $\alpha_j^{rs} = 1$, the corresponding factor in the first product is interpreted as $0^0 = 1$. The ratio Γ_{rs} measures the gain from treating rows r, s as one pair rather than as two singleton rows.

Theorem 6 (Paired lower certificate). *For every matching M on the row set—that is, every collection of disjoint unordered pairs of rows—*

$$L_M(A, X) := e^{\beta A(X)} \prod_{\{r,s\} \in M} \Gamma_{rs}(A, X) \leq \text{per}(A). \quad (10)$$

The theorem has two important features. It is one-sided for every matching, so the pairs may be chosen algorithmically, and it reduces the rest of the paper to finding many pairs with $\Gamma_{rs} > 1$. The proof encodes the row clusters and the requirement of using every column once by two stable polynomials whose coefficient inner product is the permanent.

Proof. The cluster polynomial. Let $\mathcal{G}$ be the partition of the row set whose nontrivial parts are the pairs in M . Give each cluster $C \in \mathcal{G}$ its own copy $z_{C1}, \dots, z_{Cn}$ of the column variables. For a singleton $C = \{i\}$, put

$$p_C(z_C) = \sum_j A_{ij} z_{Cj}.$$

For a pair $C = \{r, s\}$, put $p_C(z_C) = Q_{rs}(z_C)$, and define

$$p(z) = \prod_{C \in \mathcal{G}} p_C(z_C). \quad (11)$$

Stability. The singleton polynomials are positive linear forms and are therefore real stable. For a pair, write $u = A_r$ and $v = A_s$, viewed as row vectors, and let $u \circ v$ denote their coordinatewise product. The Hessian of Q_{rs} is

$$\nabla^2 Q_{rs} = uv^\top + vu^\top - 2\text{diag}(u \circ v). \quad (12)$$

Multiplying this matrix on the left and right by $\text{diag}(u \circ v)^{-1/2}$ transforms it into

$$ab^\top + ba^\top - 2I, \quad a_j = \sqrt{u_j/v_j}, \quad b_j = \sqrt{v_j/u_j}.$$

Here I is the identity matrix. The two possibly nonzero eigenvalues of $ab^\top + ba^\top$ are

$$a^\top b + \|a\| \|b\| \quad \text{and} \quad a^\top b - \|a\| \|b\|.$$

The second is nonpositive by Cauchy–Schwarz. After subtracting $2I$, the matrix therefore has at most one positive eigenvalue. The multiplication above preserves the numbers of positive and

negative eigenvalues, so the same conclusion holds for $\nabla^2 Q_{rs}$. By [Lemma 3](#), Q_{rs} is real stable. Products in disjoint variables preserve stability, so p is real stable.

The column selector. Define

$$R(z) = \prod_{j=1}^n \left(\sum_{C \in \mathcal{G}} z_{Cj} \right). \quad (13)$$

It is a product of positive linear forms and is therefore multiaffine and real stable. Both p and R are homogeneous of degree n . Their coefficient inner product is exactly the permanent:

$$\langle p, R \rangle = \text{per}(A). \quad (14)$$

Indeed, a common squarefree monomial chooses one cluster for every column and assigns to each cluster as many columns as it has rows. A singleton has one possible internal assignment. The two terms in the coefficient of $z_{Cj}z_{Ck}$ for a pair $C = \{r, s\}$ enumerate the two bijections from its rows to its selected columns. Thus the common monomials and their coefficients enumerate all permutations, with their correct A -weights.

Applying the coefficient inequality. Define a vector indexed by cluster-column pairs by

$$\alpha_{\{i\},j} = X_{ij}, \quad \alpha_{\{r,s\},j} = X_{rj} + X_{sj}. \quad (15)$$

For each cluster C , the sum of α_{Cj} over j equals the degree of p_C . For each column j , $\sum_C \alpha_{Cj} = 1$. We may therefore apply [Theorem 4](#) to p and R .

Capacities factor across disjoint variable sets. For a singleton cluster, the weighted arithmetic-geometric mean inequality (AM–GM) gives

$$\text{Cap}_{X_i} \left(\sum_j A_{ij} z_j \right) = \prod_j \left(\frac{A_{ij}}{X_{ij}} \right)^{X_{ij}}. \quad (16)$$

For a fixed column j , another application of weighted AM–GM gives

$$\text{Cap}_{(\alpha_{Cj})_C} \left(\sum_C z_{Cj} \right) = \prod_C \alpha_{Cj}^{-\alpha_{Cj}}.$$

Consequently

$$\text{Cap}_\alpha(R) = \prod_{C,j} \alpha_{Cj}^{-\alpha_{Cj}}. \quad (17)$$

The factor $\prod_{C,j} \alpha_{Cj}^{\alpha_{Cj}}$ in [Eq. \(4\)](#) cancels [Eq. \(17\)](#). What remains from a singleton cluster is exactly $S_i(A, X)$, by [Eqs. \(6\)](#) and [\(16\)](#); what remains from a pair is exactly $S_{rs}^{\text{pair}}(A, X)$. Combining [Eqs. \(4\)](#) and [\(14\)](#) yields

$$\text{per}(A) \geq \prod_{\{i\} \in \mathcal{G}} S_i(A, X) \prod_{\{r,s\} \in \mathcal{M}} S_{rs}^{\text{pair}}(A, X).$$

Using [Eqs. \(7\)](#) and [\(9\)](#) gives [Eq. \(10\)](#). □

4 An exact identity for the Bethe slack

We now ask what a matrix must look like when the upper bound in Eq. (2) is nearly tight. The starting point is a probability distribution on the permutations counted by the permanent. Comparing this distribution with a sequential sampling procedure recovers the Anari–Rezaei upper bound; keeping the full comparison gives the additional structural information that we need.

Let

$$\mu(\sigma) = \frac{\prod_i A_{i,\sigma(i)}}{\text{per}(A)} \quad (18)$$

be the Gibbs distribution on permutations, and let

$$P_{ij} = \mathbb{P}_{\sigma \sim \mu}[\sigma(i) = j]$$

be its matrix of assignment marginals: P_{ij} is the probability that row i is assigned to column j . Since A is positive, μ has full support and P is an interior point of $\mathcal{B}_n$.

For a probability vector p , define

$$T(p) = \mathbb{E}_\pi \left[\sum_j p_j \log \left(\sum_{k \geq_\pi j} p_k \right) \right], \quad (19)$$

$$g(p) = T(p) - \sum_j (1 - p_j) \log(1 - p_j), \quad (20)$$

$$d(p) = \frac{1}{2} \log 2 - g(p), \quad (21)$$

where π is a uniformly random ordering of the coordinates, and $k \geq_\pi j$ means that coordinate k appears at or after coordinate j in this ordering. Thus $T(p)$ averages the log suffix mass seen from a sample drawn from p . The quantity $g(p)$ is the resulting row correction in the sequential comparison, and $d(p)$ measures its deficit from the sharp one-row bound

$$d(p) \geq 0 \quad \text{for every probability vector } p \quad (22)$$

[AR24].

For an ordering $\pi = (i_1, \dots, i_n)$ of the rows, define a sequential distribution ν_π on permutations as follows. Process the rows in the order π . When row i_t is processed, choose an unused column j with probability proportional to $P_{i_t j}$. Let

$$D = \mathbb{E}_\pi [D_{\text{KL}}(\mu \| \nu_\pi)]. \quad (23)$$

Using only the inequality $D \geq 0$ is one of the steps that yields the Bethe upper bound. Here we retain the value of D exactly.

Lemma 7 (Exact sequential identity). *We have*

$$\log \text{per}(A) = \beta_A(P) + \sum_i g(P_i) - D. \quad (24)$$

Proof. Fix a row ordering π and a permutation σ . When row i is processed, the unused columns are precisely the columns assigned by σ to rows that appear at or after i in π . Thus

$$\log v_\pi(\sigma) = \sum_i \log P_{i,\sigma(i)} - \sum_i \log \left(\sum_{k: \sigma^{-1}(k) \text{ is at or after } i \text{ in } \pi} P_{ik} \right).$$

On the other hand,

$$\mathbb{E}_\mu[\log \mu(\sigma)] = \sum_{i,j} P_{ij} \log A_{ij} - \log \text{per}(A).$$

For fixed σ , a uniformly random row ordering induces a uniformly random ordering of the columns through σ . Averaging the preceding display over π and then over $\sigma \sim \mu$ therefore gives

$$D = \sum_{i,j} P_{ij} \log A_{ij} - \log \text{per}(A) - \sum_{i,j} P_{ij} \log P_{ij} + \sum_i T(P_i).$$

Rearranging, and using

$$T(P_i) = g(P_i) + \sum_j (1 - P_{ij}) \log(1 - P_{ij}),$$

gives [Eq. \(24\)](#). □

Define the slack in the Bethe upper bound and the Bethe suboptimality of P by

$$\Delta(A) = \frac{n}{2} \log 2 + \log \text{Bethe}(A) - \log \text{per}(A), \quad E = \log \text{Bethe}(A) - \beta_A(P). \quad (25)$$

Thus $\Delta(A)$ is the gap in the upper half of the Bethe sandwich, while E is the loss from evaluating the Bethe objective at P rather than at its maximizer.

Lemma 8 (Exact slack decomposition). *We have*

$$\Delta(A) = E + \sum_i d(P_i) + D. \quad (26)$$

In particular, every term on the right is nonnegative.

Proof. Substitute [Eq. \(24\)](#) into [Eq. \(25\)](#) and use $d(P_i) = \frac{1}{2} \log 2 - g(P_i)$. The nonnegativity of E follows from the definition of the Bethe optimum, the nonnegativity of the row deficits is [Eq. \(22\)](#), and $D \geq 0$ is Gibbs' inequality. □

5 Dimension-free stability of the row inequality

To use the slack decomposition, we need a dimension-free description of rows with small deficit. The equality examples in the row inequality are the half-half vectors. For $m \geq 2$, let

$$\mathcal{E}_m = \left\{ \frac{1}{2} e_a + \frac{1}{2} e_b \mid a, b \in [m], a \neq b \right\}. \quad (27)$$

Here e_a is the a -th standard basis vector, and

$$\text{dist}_1(p, \mathcal{E}_m) = \min_{q \in \mathcal{E}_m} \sum_j |p_j - q_j|.$$

The next lemma gives a modulus that is independent of the ambient dimension.

Lemma 9 (Row stability). *There is a universal constant $C_{\text{stab}} < \infty$ such that every probability vector $p \in \mathbb{R}_{\geq 0}^m$, $m \geq 2$, satisfies*

$$\text{dist}_1(p, \mathcal{E}_m) \leq C_{\text{stab}} d(p)^{1/4}. \quad (28)$$

Proof. The proof has two steps. First we lower-bound $d(p)$ by a separable function of the coordinates and identify its equality cases. We then turn this lower bound into a quantitative estimate on the two largest coordinates.

A separable lower bound. For a random ordering π , put

$$L_i = \sum_{j \leq \pi i} p_j, \quad R_i = \sum_{j \geq \pi i} p_j.$$

Pairing every ordering with its reverse gives

$$g(p) = \frac{1}{2} \sum_i p_i \mathbb{E}_\pi [\log(L_i R_i)] - \sum_i (1 - p_i) \log(1 - p_i). \quad (29)$$

Let $s_k = \sum_i p_i^k$. Averaging over the relative positions of three indices gives

$$M_i := \mathbb{E}_\pi [L_i R_i] = \frac{1 - s_2}{6} + \frac{2}{3} p_i + \frac{1}{3} p_i^2. \quad (30)$$

To see this, write

$$L_i R_i = p_i + \left(\sum_{j < \pi i} p_j \right) \left(\sum_{k > \pi i} p_k \right).$$

For distinct $j, k \neq i$, the probability that $j < \pi i < \pi k$ is $1/6$. Summing the corresponding cross terms proves [Eq. \(30\)](#).

Jensen's inequality and the tangent inequality

$$\log x \leq -\log 2 + 2x - 1$$

give

$$\begin{aligned} g(p) &\leq -\frac{1}{2} \log 2 + \sum_i p_i M_i - \frac{1}{2} - \sum_i (1 - p_i) \log(1 - p_i) \\ &= -\frac{1}{2} \log 2 - \frac{1}{3} + \frac{1}{2} s_2 + \frac{1}{3} s_3 - \sum_i (1 - p_i) \log(1 - p_i). \end{aligned}$$

Consequently

$$d(p) \geq C - \sum_i F(p_i), \quad (31)$$

where

$$F(x) = -(1 - x) \log(1 - x) + \frac{x^2}{2} + \frac{x^3}{3}, \quad C = \log 2 + \frac{1}{3}. \quad (32)$$

The equality case. Put $h(x) = F(x)/x$ for $x > 0$, with $h(0) = 1$. Direct differentiation gives

$$F''(x) = \frac{x(1 - 2x)}{1 - x}, \quad h'(x) = \frac{1}{x^2} \int_0^x t F''(t) dt.$$

Thus h is strictly increasing on $[0, 1/2]$, and a direct substitution gives $h(1/2) = C$.

Suppose first that every coordinate of p is at most $1/2$. Then

$$\sum_i F(p_i) = \sum_i p_i h(p_i) \leq C.$$

Equality is possible only if all positive coordinates equal $1/2$, hence only when $p \in \mathcal{E}_m$.

Now suppose that the largest coordinate is $a > 1/2$, and put $q = 1 - a$. Every remaining coordinate is at most $q \leq 1/2$, so monotonicity of h gives

$$\sum_{i:p_i \neq a} F(p_i) \leq F(q).$$

For

$$J_{\text{row}}(a) = C - F(a) - F(1 - a)$$

and $u = 2a - 1$, we have

$$J'_{\text{row}}(a) = \log \frac{a}{1-a} - 4a + 2 = 2(\text{artanh } u - u) \geq 0. \quad (33)$$

Here artanh denotes the inverse hyperbolic tangent. Moreover,

$$J_{\text{row}}(a) = \int_0^{2a-1} (\text{artanh } t - t) dt \geq \frac{(2a-1)^4}{12} = \frac{4}{3} \left(a - \frac{1}{2}\right)^4. \quad (34)$$

Thus the separable defect is nonnegative, vanishes only at a half-half vector, and grows quartically when one coordinate exceeds $1/2$.

Quantitative stability. Let $a \geq b$ be the two largest coordinates of p , and put $t = 1 - a - b$. Define

$$c_1 = \min_{x \in [1/4, 1/2]} h'(x) > 0, \quad c_2 = \min_{x \in [1/4, 1/2]} (h(x) - h(x/2)) > 0,$$

and

$$\delta_\star = \min \left\{ \frac{1}{192}, \frac{1}{2} (C - h(1/4)) \right\} > 0. \quad (35)$$

All three constants are absolute.

Assume first that $a \leq 1/2$ and $d(p) \leq \delta_\star$. Since the separable defect in [Eq. \(31\)](#) is at most $d(p)$, we must have $a, b \geq 1/4$: if $a < 1/4$, then the defect is at least $C - h(1/4)$; if $b < 1/4$, the mass outside the largest coordinate is at least $1/2$, and its contribution to the defect is at least $\frac{1}{2}(C - h(1/4))$. Monotonicity on $[1/4, 1/2]$ now gives

$$\frac{1}{2} - a \leq \frac{4d(p)}{c_1}, \quad \frac{1}{2} - b \leq \frac{4d(p)}{c_1}.$$

Indeed, the coordinates a and b contribute at least $ac_1(1/2 - a)$ and $bc_1(1/2 - b)$, respectively, to the separable defect, and both weights are at least $1/4$. Since $t = (1/2 - a) + (1/2 - b)$,

$$\text{dist}_1(p, \mathcal{E}_m) = 2t \leq \frac{16d(p)}{c_1}. \quad (36)$$

Suppose next that $a > 1/2$ and $d(p) \leq \delta_\star$. By Eq. (34),

$$a - \frac{1}{2} \leq \left(\frac{3d(p)}{4} \right)^{1/4} \leq \frac{1}{4},$$

so $q = 1 - a \geq 1/4$. Write the coordinates other than a as $x_1, \dots, x_r$, with $b = \max_\ell x_\ell$. The remaining separable defect is

$$F(q) - \sum_\ell F(x_\ell) = \sum_\ell x_\ell (h(q) - h(x_\ell)). \quad (37)$$

If $t = q - b \leq q/2$, all coordinates other than b are at most $q/2$, and their total mass is t . If $t > q/2$, then every tail coordinate is at most $q/2$, and their total mass q is at least t . In both cases Eq. (37) is at least $c_2 t$. Therefore

$$\text{dist}_1(p, \mathcal{E}_m) = 2 \left(a - \frac{1}{2} \right) + 2t \leq 2 \left(\frac{3d(p)}{4} \right)^{1/4} + \frac{2d(p)}{c_2}. \quad (38)$$

For $d(p) > \delta_\star$, use the bound $\text{dist}_1(p, \mathcal{E}_m) \leq 2$. Thus Eq. (28) holds with, for example,

$$C_{\text{stab}} = \max \left\{ \frac{16}{c_1}, 2 \left(\frac{3}{4} \right)^{1/4} + \frac{2}{c_2}, 2\delta_\star^{-1/4} \right\}.$$

□

6 Robust four-cycle structure

Row stability makes most rows close to half-half. We encode their two large coordinates as heavy edges. The column constraints give a graph of maximum degree two, and the sequential divergence controls the rows lying on long cycles.

The key observation is entropic. Once we know which rows leave the heavy-edge graph, each cycle carries at most one bit of matching ambiguity. A good row, on the other hand, contributes almost half a bit to the entropy calculation below. Cycles with at least three rows therefore create a definite surplus, which must be paid for by the divergence D . Since our logarithms are natural, one bit here means $\log 2$ units of entropy.

Fix $0 < \eta \leq 1/10$. Call row i *good* if

$$\text{dist}_1(P_i, \mathcal{E}_n) \leq \eta,$$

and let b be the number of bad rows. By Lemmas 8 and 9, every bad row has deficit at least

$$d_0 = \left(\frac{\eta}{C_{\text{stab}}} \right)^4,$$

and therefore

$$b \leq \frac{\Delta(A)}{d_0}. \quad (39)$$

For every good row, choose two coordinates witnessing its distance from $\mathcal{E}_n$. If their masses are u, v and the remaining mass is q , then

$$|u - \frac{1}{2}| + |v - \frac{1}{2}| + q \leq \eta. \quad (40)$$

In particular, both distinguished entries are at least $1/2 - \eta$, and the mass outside them is at most η .

6.1 The heavy-edge graph

Form the bipartite graph

$$F_\eta = \left\{ (i, j) \in [n] \times [n] \mid P_{ij} \geq \frac{1}{2} - \eta \right\}. \quad (41)$$

Its two vertex sets are the rows and columns of A . Every row and column has degree at most two, since $3(1/2 - \eta) > 1$, and every good row has degree exactly two.

We complete F_η to a spanning bipartite two-regular multigraph K . This can be done without deleting an edge: attach $2 - \deg(v)$ stubs to every row and column vertex, and pair the row stubs arbitrarily with the column stubs. The two sides have the same number of stubs because they have the same number of vertices and the same total degree in F_η . Parallel edges are allowed. Every component of K is therefore an even cycle, where a pair of parallel edges is allowed as a cycle containing one row. Such a doubled edge contains no good row, since a good row already has two distinct neighbors in F_η .

Let $N_K(i)$ denote the set of column neighbors of row i in K , with parallel edges ignored. Let N be the number of good rows in components of K containing at least three rows.

6.2 An entropy encoding

View a sampled permutation $\sigma \sim \mu$ as a perfect matching between the row and column vertices. Define

$$Y_i = \begin{cases} \star, & \sigma(i) \in N_K(i), \\ \sigma(i), & \sigma(i) \notin N_K(i). \end{cases}$$

Thus Y_i records the exact column used by row i when its matching edge is not in K , and otherwise only records that it uses one of its neighbors in K . Let $m(K)$ be the number of components of K containing at least two rows. The next lemma formalizes the fact that each surviving cycle costs at most one bit.

Lemma 10 (Core encoding). *We have*

$$H(\mu) \leq \sum_i H(Y_i) + m(K) \log 2. \quad (42)$$

Proof. Fix a value y of the vector $Y = (Y_1, \dots, Y_n)$. It determines the set of escaped rows and the columns assigned to them. Delete these rows and columns from K . Every remaining assignment consistent with y is a perfect matching of the resulting subgraph.

Within an original component of K , deleting vertices either leaves the whole cycle intact or breaks it into paths. An intact cycle containing at least two rows has two perfect matchings. Every path that has a perfect matching has a unique one, obtained greedily from an endpoint. A doubled edge represents only one row-column assignment. Consequently there are at most $2^{m(K)}$ permutations consistent with y , and

$$H(\sigma \mid Y) \leq m(K) \log 2.$$

Now use

$$H(\mu) = H(\sigma) \leq H(Y) + H(\sigma \mid Y) \leq \sum_i H(Y_i) + m(K) \log 2.$$

□

The KL term from Eq. (23) has a useful entropy form. For a probability vector p , put

$$s(p) = H(p) + T(p). \quad (43)$$

Lemma 11 (Entropy form of the sequential divergence). *We have*

$$D = -H(\mu) + \sum_i s(P_i). \quad (44)$$

Proof. The Gibbs law gives

$$H(\mu) = \log \text{per}(A) - \sum_{i,j} P_{ij} \log A_{ij}.$$

Substitute Eq. (24) and expand the Bethe objective. For a row p , the terms that remain are

$$- \sum_j p_j \log p_j + \sum_j (1 - p_j) \log(1 - p_j) + g(p) = H(p) + T(p).$$

Summing over the rows proves the identity. $\square$

6.3 The score of a good row

A good row contributes almost $(\log 2)/2$ beyond the information already contained in Y_i . The following Riemann-sum identity makes the dependence on the leakage explicit.

Lemma 12 (Suffix Riemann-sum identity). *Fix an ordering of a probability vector p , write its coordinates in that order as $p_1, \dots, p_m$, and put*

$$z_r = \sum_{k=r}^m p_k, \quad z_{m+1} = 0.$$

For

$$e(a, x) = a - x \log(1 + a/x), \quad e(a, 0) = a, \quad (45)$$

we have

$$\sum_{r=1}^m p_r \log z_r = -1 + \sum_{r=1}^m e(p_r, z_{r+1}). \quad (46)$$

Moreover, $e(a, x) \geq 0$, and for fixed $a \geq 0$, the function is nonincreasing in $x \geq 0$.

Proof. On the interval $[z_{r+1}, z_r]$,

$$\begin{aligned} p_r \log z_r - \int_{z_{r+1}}^{z_r} \log x \, dx &= p_r - z_{r+1} \log \left(1 + \frac{p_r}{z_{r+1}} \right) \\ &= e(p_r, z_{r+1}). \end{aligned}$$

Summing over r , and using $\int_0^1 \log x \, dx = -1$, proves Eq. (46). The inequality $\log(1 + t) \leq t$ gives $e(a, x) \geq 0$. Finally,

$$\frac{\partial}{\partial x} e(a, x) = -\log(1 + a/x) + \frac{a}{a + x} \leq 0,$$

where the last inequality follows from $\log(1 + t) \geq t/(1 + t)$. $\square$

Consider one good row. Call its two distinguished masses u, v , call the remaining mass $q = 1 - u - v$, and refer to the two distinguished coordinates as its core coordinates. The random variable Y_i merges the two core outcomes and leaves every outside outcome distinct. Hence

$$H(P_i) - H(Y_i) = (1 - q)H_2\left(\frac{u}{1 - q}\right). \quad (47)$$

In Eq. (46), keep only the two nonnegative summands corresponding to the core coordinates. With probability $1/2$, the other core coordinate precedes u ; then the suffix mass after u is at most q . With probability $1/2$, it follows u ; then that suffix mass is at most $v + q$. Monotonicity of e gives the analogous two estimates for v . Averaging over the ordering yields

$$\begin{aligned} s(P_i) - H(Y_i) &\geq \Psi(u, v, q), \\ \Psi(u, v, q) &= (1 - q)H_2\left(\frac{u}{1 - q}\right) - 1 \\ &\quad + \frac{1}{2} [e(u, q) + e(u, v + q) + e(v, q) + e(v, u + q)]. \end{aligned} \quad (48)$$

At $(u, v, q) = (1/2, 1/2, 0)$, the right side equals $(\log 2)/2$. Continuity makes this estimate uniform over all good rows.

Lemma 13 (Good-row score). *There is a nondecreasing function $\omega : [0, 1/10] \rightarrow \mathbb{R}_{\geq 0}$, with $\omega(\eta) \rightarrow 0$ as $\eta \downarrow 0$, such that every good row satisfies*

$$s(P_i) - H(Y_i) \geq \frac{1}{2} \log 2 - \omega(\eta). \quad (49)$$

Proof. The function e is continuous on $\mathbb{R}_{\geq 0}^2$ under the convention $e(a, 0) = a$. Hence Ψ is continuous in a neighborhood of $(1/2, 1/2, 0)$, where

$$\Psi(1/2, 1/2, 0) = \frac{1}{2} \log 2.$$

For $0 \leq \eta \leq 1/10$, let $\mathcal{D}_\eta$ be the compact set of triples $(u, v, q) \in \mathbb{R}_{\geq 0}^3$ satisfying

$$u + v + q = 1, \quad |u - \frac{1}{2}| + |v - \frac{1}{2}| + q \leq \eta,$$

and define

$$\omega(\eta) = \max_{(u, v, q) \in \mathcal{D}_\eta} \left(\frac{1}{2} \log 2 - \Psi(u, v, q) \right)_+.$$

Here $(x)_+ = \max\{0, x\}$. These sets are nested and shrink to $\{(1/2, 1/2, 0)\}$, so continuity implies that ω is nondecreasing and tends to zero at the origin. The claim follows from Eqs. (40) and (48). $\square$

For a bad row, the coarser estimate suffices. By Lemma 12, $T(p) \geq -1$ for every probability vector p . Since Y_i is a coarsening of an outcome sampled from P_i , $H(P_i) \geq H(Y_i)$. Therefore

$$s(P_i) - H(Y_i) \geq -1 \quad \text{for every bad row } i. \quad (50)$$

6.4 Excluding long components

We can now compare the half-bit contribution of each good row with the one-bit ambiguity of each cycle.

Lemma 14 (Robust cycle information). *For the fixed threshold η , let b be the number of bad rows and let N be the number of good rows in components of K containing at least three rows. Then*

$$D \geq \frac{\log 2}{6} N - \left(1 + \frac{\log 2}{2}\right) b - n\omega(\eta). \quad (51)$$

Proof. Let G be the set of good rows. Combining Eqs. (42), (44), (49) and (50) gives

$$D \geq |G| \left(\frac{1}{2} \log 2 - \omega(\eta) \right) - b - m(K) \log 2. \quad (52)$$

For a component C of K , let k_C, g_C, b_C be its number of rows, good rows, and bad rows. A one-row doubled component has $g_C = 0$. If $k_C = 2$, then

$$\frac{g_C}{2} - 1 = -\frac{b_C}{2}.$$

If $k_C \geq 3$, then

$$\frac{g_C}{2} - 1 \geq \frac{g_C}{6} - \frac{b_C}{3},$$

because the difference between the two sides is $(k_C - 3)/3$. Summing over the components, including the one-row components, gives

$$\frac{|G|}{2} - m(K) \geq \frac{N}{6} - \frac{b}{2}. \quad (53)$$

Substitute Eq. (53) into Eq. (52), and use $|G| \leq n$. $\square$

A component containing exactly two rows is a $K_{2,2}$. If such a component contains a bad row, it contains at most one good row, so contaminated two-row components account for at most b good rows. The number of rows in two-row components whose two rows are good is therefore at least

$$n - 2b - N. \quad (54)$$

Equivalently, K contains at least $(n - 2b - N)/2$ vertex-disjoint *clean pairs*: pairs of good rows forming a $K_{2,2}$ component.

7 Transfer to a computable Bethe point

The structural conclusions of Section 6 concern the true marginal matrix P , which is not available to the algorithm. We transfer these conclusions to a point obtained by concave optimization.

The entropy regularizer serves two purposes: it places the optimizer in the interior of the Birkhoff polytope, and its first-order optimality equations expose coordinates in which the paired gain factorizes. The main identity in this section then measures how far those coordinates can move on the heavy edges of P .

Fix a rational constant $\xi > 0$. For $n \geq 2$, let

$$\ell_n = \lceil \log_2 n \rceil, \quad \tau = \frac{\xi}{4\ell_n}. \quad (55)$$

Define the regularized Bethe objective

$$\Phi_\tau(Z) = \beta_A(Z) + \tau \sum_i H(Z_i), \quad Z \in \mathcal{B}_n, \quad (56)$$

and let X be its maximizer.

Lemma 15 (Regularized optimizer). *The maximizer X is unique and lies in the interior of $\mathcal{B}_n$. Moreover,*

$$\beta_A(X) \geq \log \text{Bethe}(A) - \xi n. \quad (57)$$

There are positive row and column scalings r_i, c_j such that

$$A_{ij} = r_i c_j X_{ij}^{1+\tau} (1 - X_{ij}) \quad \forall i, j. \quad (58)$$

Proof. The Bethe objective is concave on $\mathcal{B}_n$, and Shannon entropy is strictly concave. Thus Φ_τ is strictly concave.

To prove interiority, suppose that $Z \in \mathcal{B}_n$ is on the boundary. Let $W = J/n$ and put $Z(t) = (1-t)Z + tW$. A boundary point has a zero entry, say $Z_{ij} = 0$. Concavity of the Bethe objective gives

$$\beta_A(Z(t)) \geq (1-t)\beta_A(Z) + t\beta_A(W).$$

Coordinatewise concavity of $x \mapsto -x \log x$ gives the analogous bound for the row entropies. At the chosen zero entry, the difference from this chord is exactly

$$-\frac{t}{n} \log \frac{t}{n} - t \left(-\frac{1}{n} \log \frac{1}{n} \right) = \frac{t}{n} \log \frac{1}{t}.$$

Consequently

$$\Phi_\tau(Z(t)) - \Phi_\tau(Z) \geq t(\Phi_\tau(W) - \Phi_\tau(Z)) + \frac{\tau t}{n} \log \frac{1}{t}.$$

The last term dominates the first as $t \downarrow 0$. Thus every boundary point can be improved, and the maximizer is interior.

Since $\sum_i H(Z_i) \leq n \log n$ and $\log n \leq \ell_n \log 2$, the regularizer is at most ξn . Comparing X with a Bethe maximizer gives [Eq. \(57\)](#).

At the interior maximizer, the Karush–Kuhn–Tucker (KKT) conditions apply. The derivative of the objective in coordinate (i, j) is

$$\log A_{ij} - (1 + \tau) \log X_{ij} - \log(1 - X_{ij}) - (2 + \tau).$$

The KKT equations say that this matrix is a sum of a row potential and a column potential. Exponentiating, and absorbing the constant $2 + \tau$ into the potentials, gives [Eq. \(58\)](#). $\square$

For a row i , put

$$q_i = \prod_k (1 - X_{ik}), \quad U_{ij} = \frac{X_{ij}^{1+\tau}(1 - X_{ij})}{q_i} = \frac{X_{ij}^{1+\tau}}{\prod_{k \neq j} (1 - X_{ik})}. \quad (59)$$

The normalization by q_i makes the variables U_{ij} both bounded and compatible with the KKT factorization. These are the two properties used below.

Lemma 16 (Elementary bounds on U). *For every row i ,*

$$0 < U_{ij} \leq 1, \quad \sum_j U_{ij} \leq e. \quad (60)$$

Proof. Suppress the row index and write $x_j = X_{ij}$ and $q = \prod_k (1 - x_k)$. The elementary product inequality gives

$$\prod_{k \neq j} (1 - x_k) \geq 1 - \sum_{k \neq j} x_k = x_j.$$

Therefore

$$U_j \leq x_j^\tau \leq 1.$$

For the row sum, first discard $x_j^\tau \leq 1$:

$$\sum_j U_j \leq \frac{\sum_j x_j(1 - x_j)}{\prod_k (1 - x_k)} = \frac{1 - s_2}{q}, \quad s_2 = \sum_j x_j^2. \quad (61)$$

Put $a = \sqrt{s_2} < 1$. Monotonicity of ℓ_p -norms gives $\sum_j x_j^k \leq a^k$ for $k \geq 2$. Hence

$$\begin{aligned} -\log q &= 1 + \sum_{k \geq 2} \frac{\sum_j x_j^k}{k} \\ &\leq 1 + \sum_{k \geq 2} \frac{a^k}{k} \\ &= 1 - \log(1 - a) - a \\ &\leq 1 - \log(1 - a^2), \end{aligned}$$

where the last inequality is equivalent to $\log(1 + a) \leq a$. Thus $q \geq e^{-1}(1 - s_2)$, and [Eq. \(61\)](#) proves the result. $\square$

7.1 The transfer identity

The transfer identity is the bridge between the structural point P and the computable point X . It assigns a nonnegative cost to every entry of U . We first bound the total cost and then subtract the part caused by assignments outside K , leaving a bound on the heavy edges.

For a probability vector p , define its row contribution to the Bethe entropy by

$$h_B(p) = H(p) + \sum_j (1 - p_j) \log(1 - p_j), \quad (62)$$

and let

$$E_\tau = \Phi_\tau(X) - \Phi_\tau(P) \geq 0.$$

Thus E_τ is the suboptimality of the marginal matrix P in the regularized program.

Lemma 17 (Global transfer identity). *We have the exact identity*

$$E_\tau + \sum_i [h_B(P_i) + \tau H(P_i)] = \sum_{i,j} P_{ij} \log \frac{1}{U_{ij}}. \quad (63)$$

Moreover,

$$\sum_{i,j} P_{ij} \log \frac{1}{U_{ij}} \leq \Delta(A) + 2\xi n + H(\mu) - \frac{n}{2} \log 2. \quad (64)$$

Proof. Write $\log A_{ij}$ using [Eq. \(58\)](#). Row and column scaling terms cancel between $\Phi_\tau(X)$ and $\Phi_\tau(P)$. At X , all remaining terms sum to $\sum_i \log q_i$. Expanding the difference gives

$$\begin{aligned} E_\tau &= \sum_i \log q_i - (1 + \tau) \sum_{i,j} P_{ij} \log X_{ij} - \sum_{i,j} P_{ij} \log(1 - X_{ij}) \\ &\quad + (1 + \tau) \sum_{i,j} P_{ij} \log P_{ij} - \sum_{i,j} (1 - P_{ij}) \log(1 - P_{ij}). \end{aligned}$$

Adding $\sum_i [h_B(P_i) + \tau H(P_i)]$ cancels the second line. The first line is exactly $\sum_{i,j} P_{ij} \log(1/U_{ij})$, proving [Eq. \(63\)](#).

To obtain the upper bound, use that the regularizer is nonnegative and at most ξn :

$$0 \leq E_\tau \leq E + \xi n, \quad \tau \sum_i H(P_i) \leq \xi n.$$

By [Eq. \(44\)](#) and $s = g + h_B$,

$$\sum_i h_B(P_i) = H(\mu) + D - \sum_i g(P_i).$$

Finally,

$$\sum_i g(P_i) = \frac{n}{2} \log 2 - \sum_i d(P_i).$$

Use [Eq. \(26\)](#) to replace $E + D + \sum_i d(P_i)$ by $\Delta(A)$. This gives [Eq. \(64\)](#). $\square$

For each row i , let

$$\rho_i = \sum_{j \notin N_K(i)} P_{ij}. \quad (65)$$

Thus ρ_i is the probability that the matching edge of row i lies outside K . For a good row, $\rho_i \leq \eta$. We call the edges from i to $N_K(i)$ its core edges and define their transfer cost by

$$R_{\text{core}} = \sum_i \sum_{j \in N_K(i)} P_{ij} \log \frac{1}{U_{ij}}. \quad (66)$$

Lemma 18 (Excursion-entropy cancellation). *We have*

$$R_{\text{core}} \leq \Delta(A) + 2\xi n + \sum_i [H_2(\rho_i) + \rho_i]. \quad (67)$$

Consequently,

$$\frac{R_{\text{core}}}{n} \leq \frac{\Delta(A)}{n} + 2\xi + H_2(\eta) + \eta + (1 + \log 2) \frac{b}{n}. \quad (68)$$

Proof. Combine Eqs. (42) and (64) and $m(K) \leq n/2$ to obtain

$$\sum_{i,j} P_{ij} \log \frac{1}{U_{ij}} \leq \Delta(A) + 2\xi n + \sum_i H(Y_i). \quad (69)$$

Fix a row and let $O = [n] \setminus N_K(i)$ be its set of outside columns. Write $\rho = \sum_{j \in O} P_{ij}$ and $W = \sum_{j \in O} U_{ij}$. If $\rho = 0$, the desired estimate holds. Otherwise the log-sum inequality gives

$$\begin{aligned} \sum_{j \in O} P_{ij} \log \frac{1}{U_{ij}} &= \rho H((P_{ij}/\rho)_{j \in O}) + \sum_{j \in O} P_{ij} \log \frac{P_{ij}/\rho}{U_{ij}/W} - \rho \log W \\ &\geq \rho H((P_{ij}/\rho)_{j \in O}) - \rho, \end{aligned}$$

where we used $W \leq e$ from Lemma 16. Since

$$H(Y_i) = H_2(\rho) + \rho H((P_{ij}/\rho)_{j \in O}),$$

we obtain

$$\sum_{j \notin N_K(i)} P_{ij} \log \frac{1}{U_{ij}} \geq H(Y_i) - H_2(\rho_i) - \rho_i. \quad (70)$$

Subtract Eq. (70) from Eq. (69) and sum over the rows to obtain Eq. (67). For good rows use $\rho_i \leq \eta$; for bad rows use $H_2(\rho_i) + \rho_i \leq \log 2 + 1$. This gives the normalized bound. $\square$

8 A constant gain from a clean four-cycle

The KKT equations also factor the paired certificate. For a row i , put

$$\zeta_i = \prod_j X_{ij}^{\tau X_{ij}} \leq 1. \quad (71)$$

Lemma 19 (Pair factorization). *For two rows r, s , let $\alpha_j = X_{rj} + X_{sj}$. Then*

$$\Gamma_{rs}(A, X) = \frac{1}{\zeta_r \zeta_s} \prod_j (1 - \alpha_j)^{1 - \alpha_j} \text{Cap}_\alpha \left(\sum_{j < k} (U_{rj} U_{sk} + U_{rk} U_{sj}) z_j z_k \right). \quad (72)$$

Proof. Substituting Eq. (58) into the singleton factor gives

$$S_i(A, X) = r_i q_i \zeta_i \prod_j c_j^{X_{ij}}. \quad (73)$$

For the pair polynomial, Eqs. (58) and (59) give

$$Q_{rs}(z) = r_r r_s q_r q_s \sum_{j < k} c_j c_k (U_{rj} U_{sk} + U_{rk} U_{sj}) z_j z_k.$$

Rescaling the capacity variables by c_j therefore gives

$$\begin{aligned} \text{Cap}_\alpha(Q_{rs}) &= r_r r_s q_r q_s \left(\prod_j c_j^{\alpha_j} \right) \\ &\quad \cdot \text{Cap}_\alpha \left(\sum_{j < k} (U_{rj} U_{sk} + U_{rk} U_{sj}) z_j z_k \right). \end{aligned}$$

Divide by Eq. (73) for rows r, s . Since $\alpha_j = X_{rj} + X_{sj}$, the row scalings, column scalings, and the factors $q_r q_s$ cancel, leaving Eq. (72). $\square$

A clean pair gives a constant gain whenever its four core entries have small transfer cost.

Lemma 20 (Clean-pair gain). *There are rational constants $\kappa_0, \gamma_0, \xi_0 > 0$ with the following property. Suppose rows r, s and columns a, b form a clean $K_{2,2}$ component of K . If $0 < \xi \leq \xi_0$ and*

$$\sum_{i \in \{r, s\}} \sum_{j \in \{a, b\}} \log \frac{1}{U_{ij}} \leq \kappa_0, \quad (74)$$

then

$$\log \Gamma_{rs}(A, X) \geq \gamma_0. \quad (75)$$

Proof. The proof has two steps. Large core entries of U first force both rows of X to place almost all their mass on the two core columns. We then build a feasible distribution for the entropy dual of capacity which places most of its mass on the core monomial $z_a z_b$.

Let $\kappa > 0$ be a parameter to be fixed below and put $u_\kappa = e^{-\kappa}$. Assume the left side of Eq. (74) is at most κ . By Lemma 16, every summand in Eq. (74) is nonnegative. Thus all four core entries of U are at least u_κ .

Small transfer cost implies small leakage. Consider one of the two rows and write

$$x_a + x_b + t = 1, \quad t = \sum_{j \notin \{a, b\}} X_{ij}.$$

Since $\prod_{j \notin \{a, b\}} (1 - X_{ij}) \geq 1 - t$, we have

$$U_{ia} \leq \frac{x_a}{(1 - x_b)(1 - t)} = \frac{x_a}{x_a + t x_b}, \quad (76)$$

$$U_{ib} \leq \frac{x_b}{x_b + t x_a}. \quad (77)$$

The inequalities $U_{ia}, U_{ib} \geq u_\kappa$ imply

$$u_\kappa t x_b \leq (1 - u_\kappa) x_a, \quad u_\kappa t x_a \leq (1 - u_\kappa) x_b.$$

Multiplying and using $x_a, x_b > 0$ gives

$$t \leq u_\kappa^{-1} - 1 = e^\kappa - 1. \quad (78)$$

Let

$$\alpha_j = X_{rj} + X_{sj}, \quad \rho = \sum_{j \notin \{a,b\}} \alpha_j.$$

Applying [Eq. \(78\)](#) to both rows gives

$$\rho \leq \bar{\rho}_\kappa := 2(e^\kappa - 1). \quad (79)$$

Put $\delta_a = 1 - \alpha_a$ and $\delta_b = 1 - \alpha_b$. Since $\sum_j \alpha_j = 2$, we have $\delta_a + \delta_b = \rho$.

A capacity witness concentrated on the core. Consider the following probability distribution on two-element column sets:

$$\theta_{\{a,b\}} = 1 - \rho, \quad \theta_{\{a,\ell\}} = \frac{\delta_b}{\rho} \alpha_\ell, \quad \theta_{\{b,\ell\}} = \frac{\delta_a}{\rho} \alpha_\ell \quad (\ell \notin \{a,b\}). \quad (80)$$

At $\rho = 0$, use the limiting distribution concentrated on $\{a,b\}$. For $\rho > 0$, the total mass is one. Its inclusion marginal at an outside column ℓ is α_ℓ , while

$$1 - \rho + \delta_b = 1 - \delta_a = \alpha_a, \quad 1 - \rho + \delta_a = 1 - \delta_b = \alpha_b.$$

Thus its mean exponent vector is α , so θ is feasible in [Lemma 5](#).

Let

$$\tilde{Q}(z) = \sum_{j < k} (U_{rj}U_{sk} + U_{rk}U_{sj})z_jz_k$$

be the polynomial in [Eq. \(72\)](#). Its coefficient at $z_a z_b$ is at least $2u_\kappa^2$. If

$$V_\ell = U_{r\ell} + U_{s\ell},$$

then the coefficients at $z_a z_\ell$ and $z_b z_\ell$ are at least $u_\kappa V_\ell$. Also,

$$U_{i\ell} = \frac{X_{i\ell}^{1+\tau}}{\prod_{k \neq \ell} (1 - X_{ik})} \geq X_{i\ell}^{1+\tau}.$$

Convexity of $x \mapsto x^{1+\tau}$ therefore gives

$$V_\ell \geq X_{r\ell}^{1+\tau} + X_{s\ell}^{1+\tau} \geq 2^{-\tau} \alpha_\ell^{1+\tau}. \quad (81)$$

We apply [Lemma 5](#) to $\tilde{Q}$ with the feasible distribution θ . Its expected log coefficient is at least

$$(1 - \rho) \log(2u_\kappa^2) + \rho \log u_\kappa + \sum_{\ell \notin \{a,b\}} \alpha_\ell \log V_\ell. \quad (82)$$

Its entropy is

$$\begin{aligned} H(\theta) &= -(1 - \rho) \log(1 - \rho) - \sum_{\ell \notin \{a,b\}} \alpha_\ell \log \alpha_\ell \\ &\quad - \delta_a \log \frac{\delta_a}{\rho} - \delta_b \log \frac{\delta_b}{\rho}. \end{aligned} \quad (83)$$

Using Eq. (81) in Eq. (82), and then adding Eq. (83), gives

$$\begin{aligned} \log \text{Cap}_\alpha(\tilde{Q}) &\geq (1 - \rho) \log \frac{2u_\kappa^2}{1 - \rho} + \rho \log u_\kappa - \tau \rho \log 2 \\ &\quad + \tau \sum_{\ell \notin \{a, b\}} \alpha_\ell \log \alpha_\ell - \delta_a \log \frac{\delta_a}{\rho} - \delta_b \log \frac{\delta_b}{\rho}. \end{aligned} \quad (84)$$

Combining the factors. Insert the capacity bound into Eq. (72). Its core coordinates contribute $\delta_a \log \delta_a + \delta_b \log \delta_b$. These terms combine with the last line of Eq. (84) to give $\rho \log \rho$. The outside coordinates satisfy

$$(1 - \alpha_\ell) \log(1 - \alpha_\ell) \geq -\alpha_\ell.$$

Finally, $\zeta_r \zeta_s \leq 1$. We obtain

$$\begin{aligned} \log \Gamma_{rs} &\geq (1 - \rho) \log \frac{2u_\kappa^2}{1 - \rho} + \rho \log u_\kappa + \rho \log \rho - \rho \\ &\quad - \tau \left[- \sum_{\ell \notin \{a, b\}} \alpha_\ell \log \alpha_\ell + \rho \log 2 \right]. \end{aligned} \quad (85)$$

The entropy in brackets is at most $\rho \log(n/\rho) + \rho \log 2$. By Eq. (55), $\log n \leq \ell_n \log 2$, and $\rho \log(1/\rho) \leq 1/e$, its contribution after multiplication by τ is at most ξ times

$$\frac{1}{4} \left(\rho \log 2 + \frac{1}{e \ell_n} + \frac{\rho \log 2}{\ell_n} \right).$$

For all sufficiently small κ , we have $\rho \leq \bar{\rho}_\kappa < 1/10$, so this ratio is less than one.

For the first line of Eq. (85), set

$$f_\kappa(\rho) = (1 - \rho) \log \frac{2e^{-2\kappa}}{1 - \rho} - \kappa \rho + \rho \log \rho - \rho.$$

Here $0 \log 0 = 0$. Since $\bar{\rho}_\kappa = 2(e^\kappa - 1) \rightarrow 0$,

$$\min_{0 \leq \rho \leq \bar{\rho}_\kappa} f_\kappa(\rho) \longrightarrow \log 2 \quad \text{as } \kappa \downarrow 0.$$

Choose a rational $\kappa_0 > 0$ small enough that this minimum is larger than $(\log 2)/2$, and then choose rational $\xi_0, \gamma_0 > 0$ such that

$$\gamma_0 < \frac{1}{2} \log 2 - \xi_0.$$

Taking $\kappa = \kappa_0$ in the preceding argument, Eq. (85) proves the lemma for these constants. $\square$

The same witnesses make the gain algorithmic without requiring a separate capacity optimization. For distinct columns a, b whose outside mass ρ is at most one, define $\theta^{a,b}$ by Eq. (80), with the point mass on $\{a, b\}$ when $\rho = 0$, and write

$$c_{jk} = U_{rj} U_{sk} + U_{rk} U_{sj}.$$

Set

$$\begin{aligned} \log \Gamma_{rs}^{a,b} &:= -\log \zeta_r - \log \zeta_s + \sum_j (1 - \alpha_j) \log(1 - \alpha_j) \\ &\quad + \sum_{j < k} \theta_{jk}^{a,b} \log \frac{c_{jk}}{\theta_{jk}^{a,b}}. \end{aligned} \quad (86)$$

By [Lemmas 5](#) and [19](#), $\Gamma_{rs}^{a,b} \leq \Gamma_{rs}$. More importantly, the proof of [Lemma 20](#) used precisely the witness with the two core columns; it therefore proves the stronger conclusion

$$\log \Gamma_{rs}^{a,b} \geq \gamma_0 \quad (87)$$

for that choice of a, b .

For the algorithm, we assign row pair $\{r, s\}$ the nonnegative weight

$$\underline{w}_{rs} = \max \left\{ 0, \max_{a \neq b: \rho_{rs}^{a,b} \leq 1} \log \Gamma_{rs}^{a,b} \right\}. \quad (88)$$

If there is no eligible column pair, the inner maximum is omitted and the weight is zero. The core columns of every clean pair are eligible by [Eq. \(79\)](#) and the choice of κ_0 . A maximum-weight matching in the complete graph on the rows gives at least as much total gain as any collection of clean pairs constructed only for the analysis, and can be found in polynomial time [[Sch03](#)]. Edges of zero weight are omitted from the final certificate.

9 Completing the approximation bound

We now combine the stability information from the Bethe upper bound with the gain from the paired lower certificate. Let $\kappa_0, \gamma_0, \xi_0$ be the constants from [Lemma 20](#). Enlarge C_{stab} , if necessary, to make it rational.

The exact values of the constants will not matter. We choose them once, in the order used by the proof: η controls the structural errors, δ_0 defines the near-equality regime, and ξ controls the regularization loss. Choose rational $\eta, \delta_0, \xi > 0$, in this order, and put

$$d_0 = \left(\frac{\eta}{C_{\text{stab}}} \right)^4.$$

The following three quantities will bound, respectively, the fractions of bad rows, rows on long components, and clean pairs with excessive transfer cost. We require $\eta \leq 1/10$ and

$$\varepsilon_{\text{row}} := \frac{\delta_0}{d_0} \leq \frac{1}{128}, \quad (89)$$

$$\varepsilon_{\text{cyc}} := \frac{6}{\log 2} \left[\delta_0 + \left(1 + \frac{\log 2}{2} \right) \varepsilon_{\text{row}} + \omega(\eta) \right] \leq \frac{1}{16}, \quad (90)$$

$$\varepsilon_{\text{tr}} := \frac{\delta_0 + 2\xi + H_2(\eta) + \eta + (1 + \log 2)\varepsilon_{\text{row}}}{(1/2 - \eta)\kappa_0} \leq \frac{1}{16}, \quad (91)$$

$$\xi \leq \min \left\{ \xi_0, \frac{\delta_0}{2}, \frac{\gamma_0}{16} \right\}. \quad (92)$$

Such a choice exists. First take η small enough to control $\omega(\eta)$ and $H_2(\eta) + \eta$; next take δ_0 small relative to d_0 ; finally take ξ small. No numerical optimization of these constants is needed.

Let X be the regularized point from [Section 7](#). Give every row pair the weight in [Eq. \(88\)](#), and let M be a maximum-weight matching. Delete the zero-weight edges from M , and define

$$L_X(A) = \exp \left(\beta_A(X) + \sum_{\{r,s\} \in M} \frac{w_{rs}}{2} \right). \quad (93)$$

For every positive-weight edge, choose columns attaining the maximum in [Eq. \(88\)](#). Since its witness gain is no larger than Γ_{rs} , [Theorem 6](#) gives $L_X(A) \leq \text{per}(A)$.

Proposition 21 (Positive matrices). *There is an absolute constant $\varepsilon_+ > 0$ such that every positive matrix A of order $n \geq 2$ satisfies*

$$L_X(A) \leq \text{per}(A) \leq \left(\sqrt{2}e^{-\varepsilon_+} \right)^n L_X(A). \quad (94)$$

Proof. The proof is a dichotomy between slack in the Bethe bound and gain from the paired certificate. Recall that b counts bad rows, N counts good rows in long components of K , D is the averaged sequential divergence, and R_{core} is the transfer cost on the edges of K .

Case 1: nonnegligible Bethe slack. Suppose that $\Delta(A) \geq \delta_0 n$. Since the matching gain in [Eq. \(93\)](#) is nonnegative, [Eq. \(57\)](#) gives

$$\begin{aligned} \log \frac{\text{per}(A)}{L_X(A)} &\leq \log \text{per}(A) - \beta_A(X) \\ &\leq \left(\frac{1}{2} \log 2 - \delta_0 + \xi \right) n. \end{aligned} \quad (95)$$

Case 2: nearly tight Bethe bound. Suppose that $\Delta(A) < \delta_0 n$. By [Eqs. \(39\)](#) and [\(89\)](#),

$$b \leq \varepsilon_{\text{row}} n. \quad (96)$$

Also $D \leq \Delta(A)$. Rearranging [Eq. \(51\)](#) and using [Eq. \(90\)](#) gives

$$N \leq \varepsilon_{\text{cyc}} n. \quad (97)$$

Hence [Eq. \(54\)](#) gives at least

$$\frac{1 - 2\varepsilon_{\text{row}} - \varepsilon_{\text{cyc}}}{2} n \geq \frac{59}{128} n$$

vertex-disjoint clean pairs.

By [Eqs. \(68\)](#) and [\(91\)](#),

$$R_{\text{core}} \leq \varepsilon_{\text{tr}}(1/2 - \eta)\kappa_0 n. \quad (98)$$

If a clean pair with core columns a, b fails [Eq. \(74\)](#), its contribution to R_{core} is at least $(1/2 - \eta)\kappa_0$, because each of its four core P -entries is at least $1/2 - \eta$. Therefore at most $\varepsilon_{\text{tr}} n \leq n/16$ clean pairs fail the local-cost condition. At least $3n/8$ vertex-disjoint pairs satisfy the hypothesis of [Lemma 20](#). The maximum-weight matching therefore has total gain at least

$$\sum_{\{r,s\} \in M} \frac{w_{rs}}{2} \geq \frac{3\gamma_0}{8} n. \quad (99)$$

Using the upper half of the Bethe sandwich and [Eq. \(57\)](#),

$$\log \frac{\text{per}(A)}{L_X(A)} \leq \left(\frac{1}{2} \log 2 + \xi - \frac{3\gamma_0}{8} \right) n. \quad (100)$$

Define

$$\varepsilon_+ = \min \left\{ \delta_0 - \xi, \frac{3\gamma_0}{8} - \xi \right\} > 0. \quad (101)$$

[Equations \(95\)](#) and [\(100\)](#) prove [Eq. \(94\)](#). $\square$

The proposition is the conceptual core of the theorem. The remaining issues are positivity and finite-precision computation; we handle them in the appendix.

A Zero entries and finite-precision computation

This appendix passes from the exact positive-matrix argument to the algorithm in [Theorem 1](#). There are two issues: zero entries, which we remove by a one-sided smoothing, and the finite-precision computation of the regularized optimizer and the paired gains. Neither issue changes the structural argument.

A.1 Smoothing zero entries

We first extend the positive-matrix result to nonnegative matrices. We choose the smoothing to have polynomial bit complexity and to preserve a one-sided certificate.

Lemma 22 (Smoothing). *Suppose that a nonnegative rational matrix A has largest entry at most one and the bipartite graph of its positive entries has a perfect matching. There is a polynomial-time construction of a positive rational matrix $\tilde{A}$ such that*

$$\text{per}(A) \leq \text{per}(\tilde{A}) \leq \left(1 + \frac{\chi^n}{2} \right) \text{per}(A), \quad (102)$$

where $\chi > 0$ is any fixed rational constant.

Proof. Let m be the smallest positive entry of A , and put

$$\delta_{\text{sm}} = \min \left\{ \frac{1}{2n}, \frac{\chi m^n}{4n!} \right\}, \quad \tilde{A} = A + \delta_{\text{sm}} J. \quad (103)$$

Here J is the all-ones matrix. For every permutation σ ,

$$0 \leq \prod_i (A_{i,\sigma(i)} + \delta_{\text{sm}}) - \prod_i A_{i,\sigma(i)} \leq (1 + \delta_{\text{sm}})^n - 1 \leq 2n\delta_{\text{sm}}.$$

For the last inequality, use $(1 + \delta_{\text{sm}})^n \leq \sum_{k \geq 0} (n\delta_{\text{sm}})^k = (1 - n\delta_{\text{sm}})^{-1}$ and $n\delta_{\text{sm}} \leq 1/2$. Summing over the $n!$ permutations gives

$$0 \leq \text{per}(\tilde{A}) - \text{per}(A) \leq 2nn!\delta_{\text{sm}} \leq \frac{\chi^n}{2} m^n.$$

The positive support contains a perfect matching, so $\text{per}(A) \geq m^n$. This proves [Eq. \(102\)](#). The bit length of δ_{sm} is polynomial in the bit length of A and in n . $\square$

A.2 Certified finite-precision computation

It remains to compute the regularized point and the explicit witness gains with certified precision while preserving the lower-bound direction. We first bound the optimizer away from the boundary. We then solve a bounded epigraph problem by weak optimization, recover approximate KKT potentials, and interpret the result as an exact optimizer for a nearby matrix. This last step lets us reuse the exact certificate proved in the main argument.

Lemma 23 (Interior bound). *Assume $n \geq 2$ and $m \leq A_{ij} \leq 1$ for every i, j , where $m > 0$, and let X be the maximizer of Φ_τ . Then*

$$\log \frac{1}{\min_{i,j} X_{ij}} \leq \frac{nR_A}{\tau} + n^2 \log n, \quad (104)$$

where one may take

$$R_A = n \log \frac{1}{m} + n \log n + n. \quad (105)$$

In particular, if A is rational, every X_{ij} and every $1 - X_{ij}$ is at least $2^{-\text{poly}(n, B, 1/\tau)}$, where B is the input bit length.

Proof. Let J be the all-ones matrix, set $\bar{X} = J/n$, and write

$$\Phi_\tau(Z) = \beta_A(Z) + \tau \sum_i H(Z_i).$$

The range of β_A on $\mathcal{B}_n$ is at most R_A : the linear term $\sum_{i,j} Z_{ij} \log A_{ij}$ lies between $n \log m$ and zero, the row entropy is at most $n \log n$, and

$$-\sum_{i,j} Z_{ij} \leq \sum_{i,j} (1 - Z_{ij}) \log(1 - Z_{ij}) \leq 0.$$

Here the first inequality follows from $-x \leq (1 - x) \log(1 - x)$, and $\sum_{i,j} Z_{ij} = n$.

Since X is an interior maximizer, the directional derivative of Φ_τ at X toward $\bar{X}$ is zero. Concavity of β_A gives

$$\beta'_A(X; \bar{X} - X) \geq \beta_A(\bar{X}) - \beta_A(X) \geq -R_A.$$

Consequently

$$\left(\sum_i H(X_i) \right)' (X; \bar{X} - X) \leq \frac{R_A}{\tau}. \quad (106)$$

The entropy derivative is

$$-\sum_{i,j} \left(\frac{1}{n} - X_{ij} \right) (1 + \log X_{ij}) = \sum_{i,j} X_{ij} \log X_{ij} - \frac{1}{n} \sum_{i,j} \log X_{ij},$$

because the direction has coordinate sum zero. Let $x_\star = \min_{i,j} X_{ij}$. We have

$$\sum_{i,j} X_{ij} \log X_{ij} \geq -n \log n, \quad -\frac{1}{n} \sum_{i,j} \log X_{ij} \geq \frac{1}{n} \log \frac{1}{x_\star}.$$

Substitute these estimates into [Eq. \(106\)](#) to obtain [Eq. \(104\)](#). Finally,

$$1 - X_{ij} = \sum_{k \neq j} X_{ik} \geq (n - 1)x_{\star},$$

which gives the same polynomial logarithmic bound for $1 - X_{ij}$. $\square$

Lemma 24 (Certified computation). *There is an absolute constant C_{num} with the following property. For every positive rational matrix A of order $n \geq 2$ and every rational $\varepsilon_{\text{num}} > 0$, a deterministic algorithm computes a rational number $\widehat{L}$ such that*

$$\widehat{L} \leq \text{per}(A) \leq \left(\sqrt{2}e^{-\varepsilon_{\text{num}} + C_{\text{num}}\varepsilon_{\text{num}}} \right)^n \widehat{L}. \quad (107)$$

The running time is polynomial in the input bit length, n , and $\log(2 + 1/\varepsilon_{\text{num}})$.

Proof. Replacing ε_{num} by $\min\{\varepsilon_{\text{num}}, 1\}$, we may assume it is at most one. Recall that $\tau = \xi/(4\ell_n)$ is rational and that both its bit length and $1/\tau$ are polynomially bounded.

A compact optimization domain. After a global scaling, assume $m \leq A_{ij} \leq 1$, where m is the smallest entry of the scaled matrix, and let B be its input bit length. By [Lemma 23](#), the optimizer belongs to the truncated Birkhoff polytope

$$\mathcal{B}_n(\delta) = \{Z \in \mathcal{B}_n \mid Z_{ij} \geq \delta \forall i, j\}$$

for an explicitly computable rational $\delta_{\text{int}} = 2^{-\text{poly}(n, B, 1/\tau)}$. Concretely, increase B if necessary so that $m \geq 2^{-B}$, and put

$$K_0 = \frac{n(nB + 2n^2)}{\tau} + n^3, \quad q = \lceil 2K_0 \rceil, \quad \delta_{\text{int}} = 2^{-q}.$$

The inequalities $\log 2 \leq 1$ and $\log 2 \geq 1/2$ show directly from [Eq. \(104\)](#) that this floor is valid. We optimize over $\mathcal{B}_n(\delta_{\text{int}}/2)$, which still contains the exact optimizer. Since $q \geq n + 1$, $\delta_{\text{int}} \leq 1/(2n)$. On the slightly larger set $\mathcal{B}_n(\delta_{\text{int}}/4)$, both Z_{ij} and $1 - Z_{ij}$ are bounded below by $\delta_{\text{int}}/4$.

Write $f = -\Phi_{\tau}$. Its gradient is

$$\nabla f(Z)_{ij} = -\log A_{ij} + (1 + \tau) \log Z_{ij} + \log(1 - Z_{ij}) + (2 + \tau). \quad (108)$$

On the larger truncated set, its Frobenius norm is at most

$$G = n[B + (2 + \tau)(q + 3)],$$

where we used $\log 2 \leq 1$. Thus f is G -Lipschitz there. Its gradient is L -Lipschitz there, with

$$L = \frac{4(2 + \tau)}{\delta_{\text{int}}}. \quad (109)$$

Both G and $\log L$ have polynomial size.

A bounded epigraph and its separation oracle. We eliminate the row- and column-sum equations by rational affine coordinates and apply weak linear optimization to the epigraph of f . We record the geometry needed for this reduction. Use the upper-left $(n - 1)$ -by- $(n - 1)$ block as coordinates and

recover the last row and column from the sum constraints. The resulting affine map T has norm at most n from Euclidean coordinate error to Frobenius error. Also

$$|f(Z)| \leq M_0 := nB + (1 + \tau)n^2 + n \quad (Z \in \mathcal{B}_n).$$

We use the bounded epigraph

$$K = \{(y, t) \mid T(y) \in \mathcal{B}_n(\delta_{\text{int}}/2), f(T(y)) \leq t \leq M_0 + 2\}.$$

Since $\|y\|_2 \leq n$ and $-M_0 \leq t \leq M_0 + 2$ on K , it lies in the ball centered at the origin of rational radius $R = M_0 + n + 2$. If y_U represents the uniform matrix, then $(y_U, M_0 + 1)$ is the center of a ball in K of radius at least

$$r = \min \left\{ \frac{1}{4}, \frac{1}{4n^2}, \frac{1}{4nG} \right\}.$$

Indeed, changing y_U by Euclidean distance at most r changes every entry of $T(y_U)$ by at most $nr \leq 1/(4n)$, so all entries remain above $\delta_{\text{int}}/2$. Changing t by at most $r \leq 1/4$ preserves its upper bound. Finally, $f(T(y)) \leq f(T(y_U)) + Gnr \leq M_0 + 1/4$, whereas $t \geq M_0 + 3/4$, so the lower epigraph inequality also holds. Thus the outer radius and $\log(1/r)$ have polynomial bit length, exactly as required by the weak optimization theorem of [GLS81].

The weak separation oracle first checks all rational linear inequalities defining $\mathcal{B}_n(\delta_{\text{int}}/2)$ and the upper bound on t exactly. At a point passing these checks, directed approximations to Eq. (108) and to f give the usual supporting hyperplane to the epigraph, with normal $(T_{\text{lin}}^* \nabla f(T(y)), -1)$. Convexity implies that its value on the displacement from the query to any $(z, s) \in K$ satisfies

$$\langle (T_{\text{lin}}^* \nabla f(T(y)), -1), (z - y, s - t) \rangle \leq t - f(T(y)).$$

If the sign of the violation $f(T(y)) - t$ cannot be certified at the requested precision, the oracle declares the query to be in the prescribed weak enlargement. This is the weak-separation model used in the separation–optimization reduction. Taking the value error and the gradient error times the outer radius below a fixed fraction of the requested weak tolerance makes each reported cut valid. Evaluation to p directed bits takes time polynomial in n, B, q, p , so this additional accuracy is polynomially bounded.

A rounded ellipsoid implementation. We use a fixed dyadic precision for the entire ellipsoid run. This avoids evaluating the determinant of an intermediate rational matrix. To see that such a precision exists, clear the denominators of the initial rational basis. If their least common multiple is D , nonsingularity gives the explicit lower bound D^{-d} on the absolute determinant, where d is the dimension of the epigraph. Choose L so that $2^{-L} \leq D^{-d}$, and choose K so that the initial center and basis have total absolute magnitude at most 2^K . After t rounded central cuts, the determinant is at least $2^{-(L+2t)}$, while the total magnitude is at most $2^{K+t(6+3d)}$. The first estimate follows because the exact central cut retains at least half of the determinant and sufficiently fine rounding retains another half. The second follows from the entrywise central-cut and rounding bounds. It is therefore enough to use, at every iteration, the single precision

$$P = (L + 2T + 1) + 12 + 8d + d^2 + d(K + T(6 + 3d) + 3 + d) + 2,$$

where T is the iteration budget. The determinant perturbation bound and the adjugate formula show that flooring the center and basis to the 2^{-P} grid, followed by inflation by $1 + 1/(1024d^4)$, preserves the surviving half-ellipsoid. The same estimates give determinant contraction by a factor

at most $1 - 1/(32d^3)$. Thus P has polynomial length, is computed before the run from rational data, and controls every stored state.

Recovering approximate KKT potentials. Let $\Delta_{\text{obj}} > 0$ be the desired objective error. Invoke weak optimization with accuracy

$$\sigma \leq \min \left\{ \frac{\delta_{\text{int}}}{4n}, \frac{\Delta_{\text{obj}}}{2 + nG} \right\}. \quad (110)$$

Write the rational output as $(\hat{y}, \hat{t})$. It is within σ of some $(y', t') \in K$, and $\hat{t}$ is within σ above the minimum. Mapping $\hat{y}$ through T gives a rational matrix $\hat{X}$. Its row and column sums are exactly one. The first bound in Eq. (110) and the norm bound for T imply $\hat{X}_{ij} \geq \delta_{\text{int}}/4$; exact row sums then give the same lower bound for $1 - \hat{X}_{ij}$. The second bound and Lipschitzness give, more explicitly,

$$f(T(\hat{y})) \leq f(T(y')) + nG\sigma \leq t' + nG\sigma \leq \hat{t} + (1 + nG)\sigma \leq \min_K t + (2 + nG)\sigma.$$

Consequently,

$$\Phi_\tau(X) - \Phi_\tau(\hat{X}) \leq \Delta_{\text{obj}}. \quad (111)$$

Thus the fact that a weak optimizer need not return a point in its input body does not compromise exact double stochasticity.

The function f is τ -strongly convex on the affine hull of $\mathcal{B}_n$, because the negative Shannon term has Hessian $\tau \text{diag}(1/Z_{ij}) \geq \tau I$, while concavity of the Bethe objective makes the remaining Hessian positive semidefinite on the tangent space. Strong convexity gives

$$\|\hat{X} - X\|_F \leq \sqrt{2(\Phi_\tau(X) - \Phi_\tau(\hat{X}))/\tau}.$$

Set $\Delta_{\text{obj}} = \tau \varepsilon_{\text{num}}^2 / (128L^2)$ in Eq. (110). Then the two gradients are within $\varepsilon_{\text{num}}/8$ in Frobenius norm. Approximate the gradient rationally to accuracy $\varepsilon_{\text{num}}/16$. The exact optimizer's gradient has the form $(R_i + C_j)_{ij}$ by the KKT equations. Fix a row i_0 and a column j_0 , let $\tilde{G}$ be the rational gradient approximation, and set

$$\tilde{R}_i = \tilde{G}_{ij_0}, \quad \tilde{C}_j = \tilde{G}_{i_0j} - \tilde{G}_{i_0j_0}.$$

For an exact row-plus-column matrix this reconstruction is an identity. If each entry of $\tilde{G}$ is within a of such a matrix, its residual is at most $4a$, by the triangle inequality on the corresponding four-cycle. Our preceding choices make the residual, including the error in evaluating the gradient at $\hat{X}$, at most ε_{num} . Set $\lambda_i = -\tilde{R}_i$ and $\nu_j = 2 + \tau - \tilde{C}_j$. These rational potentials satisfy

$$\left| \log A_{ij} - \lambda_i - \nu_j - (1 + \tau) \log \hat{X}_{ij} - \log(1 - \hat{X}_{ij}) \right| \leq \varepsilon_{\text{num}}. \quad (112)$$

Exactification by a nearby matrix. Define the nearby positive matrix

$$A'_{ij} = e^{\lambda_i + \nu_j} \hat{X}_{ij}^{1+\tau} (1 - \hat{X}_{ij}). \quad (113)$$

Then $\hat{X}$ satisfies the exact KKT equations for A' , and concavity shows that it is the exact regularized optimizer. Moreover,

$$e^{-\varepsilon_{\text{num}}} A' \leq A \leq e^{\varepsilon_{\text{num}}} A' \quad (114)$$

entrywise. Hence

$$e^{-\varepsilon_{\text{num}} n} \text{per}(A') \leq \text{per}(A) \leq e^{\varepsilon_{\text{num}} n} \text{per}(A'). \quad (115)$$

One-sided evaluation of the paired certificate. The gain does not require another convex program. For every row pair and every pair of distinct candidate core columns, compute the outside mass and retain the candidate only if it is at most one. Then form the rational distribution $\theta^{a,b}$ in Eq. (80); the case $\rho = 0$ is detected exactly and uses the point mass on $\{a, b\}$. These are exactly the finitely many witness gains in Eq. (86). There are only $O(n^4)$ of them. Every retained $\theta^{a,b}$ satisfies its mass and marginal constraints exactly, so Lemma 5 preserves the lower-bound direction without any feasibility recovery.

Logarithms and exponentials at rational arguments can be evaluated to p directed bits in time polynomial in the input length and p [Bre76]. We use outward-rounded interval arithmetic for all elementary expressions generated from the rational inputs. There is no need to materialize the generally irrational matrix A' : inside $\beta_{A'}(\widehat{X})$ we use

$$\log A'_{ij} = \lambda_i + v_j + (1 + \tau) \log \widehat{X}_{ij} + \log(1 - \widehat{X}_{ij}),$$

and the transfer coefficients are evaluated in logarithmic form from

$$\log U_{ij} = (1 + \tau) \log \widehat{X}_{ij} - \sum_{k \neq j} \log(1 - \widehat{X}_{ik}).$$

The quantities ρ, δ_a, δ_b and every nonzero witness mass are rational. Zero masses are recognized exactly and their entropy terms are set to zero. We evaluate $\beta_{A'}(\widehat{X})$ and every witness log-gain, always taking the lower endpoint. For each row pair, take the largest lower endpoint over the candidate columns, truncate it at zero, and compute a maximum-weight matching. If every edge weight is underestimated by at most ε_{num} , the computed matching loses at most $n\varepsilon_{\text{num}}/2$ relative to the exact witness-weight matching, because both matchings have at most $n/2$ edges. Evaluating each of the polynomially many elementary terms to $\varepsilon_{\text{num}}/\text{poly}(n)$ therefore gives a rational value $\widehat{L}'$ such that

$$\widehat{L}' \leq L_{\widehat{X}}(A'), \quad \log \widehat{L}' \geq \log L_{\widehat{X}}(A') - C_0 \varepsilon_{\text{num}} n$$

for an absolute constant C_0 . All inequalities are one-sided, so $\widehat{L}' \leq \text{per}(A')$ by Theorem 6 and Eq. (86).

Assembling the output. Finally compute a rational $\widehat{L}$ satisfying

$$e^{-2\varepsilon_{\text{num}} n} \widehat{L}' \leq \widehat{L} \leq e^{-\varepsilon_{\text{num}} n} \widehat{L}'.$$

Such a number is obtained by directed evaluation with polynomially many bits. The lower bound in Eq. (107) follows from Eq. (115). Since $\widehat{X}$ is the exact regularized optimizer for A' , Proposition 21 gives

$$\text{per}(A') \leq \left(\sqrt{2} e^{-\varepsilon_+} \right)^n L_{\widehat{X}}(A').$$

Combining this inequality with Eq. (115) and the loss in $\widehat{L}'$ proves the upper bound in Eq. (107) with $C_{\text{num}} = C_0 + 3$. The initial scaling can be undone using the degree- n homogeneity of the permanent and the certificate. All requested precisions have polynomial bit length, and every rational intermediate and the final output have polynomial bit length. Rational arithmetic, weak optimization, directed elementary-function evaluation, and weighted matching therefore take polynomial time in the ordinary binary encoding of the input. $\square$

A.3 Proof of the main theorem

Proof of Theorem 1. The case $n = 1$ is exact. For $n \geq 2$, first test by bipartite matching whether the positive support of A contains a perfect matching. If not, return zero.

Otherwise scale A by its largest entry. The constant ε_+ in Eq. (101) is positive and rational. Decreasing it if necessary, we may assume that $\varepsilon_+ \leq (\log 2)/2$; this only weakens Eq. (94) and ensures that the final approximation base is larger than one. Apply Lemma 22 with $\chi = \varepsilon_+/2$, and apply Lemma 24 to the smoothed matrix with

$$\varepsilon_{\text{num}} \leq \frac{\varepsilon_+}{4C_{\text{num}}}. \quad (116)$$

By Lemma 24, the smoothed permanent is bounded by the computed certificate with base

$$\sqrt{2}e^{-\varepsilon_+ + C_{\text{num}}\varepsilon_{\text{num}}}.$$

If $\tilde{L}$ is the certified lower bound for the smoothed permanent, put

$$L = \frac{\tilde{L}}{1 + \chi^n/2}.$$

By Eq. (102), $L \leq \text{per}(A)$. On the logarithmic scale, the division loses at most

$$\log\left(1 + \frac{\chi^n}{2}\right) \leq \frac{\chi^n}{2}.$$

Our choices of χ and ε_{num} leave at least $\varepsilon_+/2$ in the exponent. Thus the final approximation base can be taken to be

$$c = \sqrt{2}e^{-\varepsilon_+/2} < \sqrt{2}. \quad (117)$$

Undoing the initial scaling multiplies both the output and the permanent by the same factor to the n th power. Every step is deterministic and polynomial-time. $\square$

B Provenance

The proof in this paper originated in an interaction between the author and ChatGPT 5.6 Sol Pro. The author supplied the high-level plan of attack and guided the development of the argument. Within this plan, the model proposed the paired stable-polynomial certificate, the exact-slack strategy, and the transfer through a regularized Bethe point. The author subsequently verified the results.

Codex assisted with alternating manuscript expansion and critical proof passes, including symbolic and numerical checks, literature verification, assembly, and typesetting. The author is responsible for the final contents.

An accompanying Lean 4 development, available at <https://github.com/nimaanari/formalization-beyond-bethe> and registered with the Palomar Registry at <https://palomar-registry.org/entry?id=PALOMAR-2026-08-29-000006&version=1>, formalizes the proof and the complete algorithm. It reconstructs from their source proofs Vontobel’s concavity theorem, the sharp one-row inequality of Anari and Rezaei, the upper Bethe bound, and the stable coefficient

inequality of Anari and Oveis Gharan. The lower Bethe bound is recovered internally as a special case.

The formalization includes the convex optimization, finite-precision error analysis, support and boundary cases, and an executable rational algorithm. It also proves ordinary deterministic polynomial running time on a binary encoding, including the elementary arithmetic, ellipsoid, matching, and certificate routines used by the algorithm. Its main theorem has no project-specific hypotheses, and its axiom audit reports only the standard logical axioms used by Lean and Mathlib.

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